```
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
import plotly.io as pio
import plotly.colors as colors
pio.templates.default="plotly_white"

data = pd.read_csv("Sample - Superstore.csv", encoding= 'latin-1')

data
```

```
      Row ID        Order ID  Order Date   Ship Date       Ship
Mode  \
0          1  CA-2016-152156   11/8/2016  11/11/2016    Second Class

1          2  CA-2016-152156   11/8/2016  11/11/2016    Second Class

2          3  CA-2016-138688   6/12/2016   6/16/2016    Second Class

3          4  US-2015-108966  10/11/2015  10/18/2015  Standard Class

4          5  US-2015-108966  10/11/2015  10/18/2015  Standard Class

...      ...             ...         ...         ...             ...

9989    9990  CA-2014-110422   1/21/2014   1/23/2014    Second Class

9990    9991  CA-2017-121258   2/26/2017    3/3/2017  Standard Class

9991    9992  CA-2017-121258   2/26/2017    3/3/2017  Standard Class

9992    9993  CA-2017-121258   2/26/2017    3/3/2017  Standard Class

9993    9994  CA-2017-119914    5/4/2017    5/9/2017    Second Class


     Customer ID     Customer Name    Segment        Country
City  \
0       CG-12520       Claire Gute   Consumer  United States
Henderson
1       CG-12520       Claire Gute   Consumer  United States
Henderson
2       DV-13045   Darrin Van Huff  Corporate  United States      Los
Angeles
3       SO-20335    Sean O'Donnell   Consumer  United States   Fort
Lauderdale
4       SO-20335    Sean O'Donnell   Consumer  United States   Fort
Lauderdale
...          ...               ...        ...            ...
...
9989    TB-21400  Tom Boeckenhauer   Consumer  United States
```

```
Miami
9990    DB-13060       Dave Brooks   Consumer  United States
Costa Mesa
9991    DB-13060       Dave Brooks   Consumer  United States
Costa Mesa
9992    DB-13060       Dave Brooks   Consumer  United States
Costa Mesa
9993    CC-12220      Chris Cortes   Consumer  United States
Westminster

      ... Postal Code  Region       Product ID         Category Sub-
Category  \
0     ...       42420   South  FUR-BO-10001798        Furniture
Bookcases
1     ...       42420   South  FUR-CH-10000454        Furniture
Chairs
2     ...       90036    West  OFF-LA-10000240  Office Supplies
Labels
3     ...       33311   South  FUR-TA-10000577        Furniture
Tables
4     ...       33311   South  OFF-ST-10000760  Office Supplies
Storage
...   ...         ...     ...              ...              ...
...
9989  ...       33180   South  FUR-FU-10001889        Furniture
Furnishings
9990  ...       92627    West  FUR-FU-10000747        Furniture
Furnishings
9991  ...       92627    West  TEC-PH-10003645       Technology
Phones
9992  ...       92627    West  OFF-PA-10004041  Office Supplies
Paper
9993  ...       92683    West  OFF-AP-10002684  Office Supplies
Appliances

                                            Product Name     Sales
Quantity  \
0                      Bush Somerset Collection Bookcase  261.9600
2
1     Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
3
2     Self-Adhesive Address Labels for Typewriters b...   14.6200
2
3         Bretford CR4500 Series Slim Rectangular Table  957.5775
5
4                         Eldon Fold 'N Roll Cart System   22.3680
2
...                                                  ...       ...
...
```

```
9989                              Ultra Door Pull Handle   25.2480
3
9990  Tenex B1-RE Series Chair Mats for Low Pile Car...   91.9600
2
9991                               Aastra 57i VoIP phone  258.5760
2
9992  It's Hot Message Books with Stickers, 2 3/4" x 5"   29.6000
4
9993  Acco 7-Outlet Masterpiece Power Center, Wihtou...  243.1600
2

      Discount    Profit
0         0.00   41.9136
1         0.00  219.5820
2         0.00    6.8714
3         0.45 -383.0310
4         0.20    2.5164
...        ...       ...
9989      0.20    4.1028
9990      0.00   15.6332
9991      0.20   19.3932
9992      0.00   13.3200
9993      0.00   72.9480

[9994 rows x 21 columns]
```

```
data.head()
```

```
   Row ID        Order ID Order Date  Ship Date       Ship Mode
Customer ID  \
0       1  CA-2016-152156 2016-11-08 2016-11-11    Second Class    CG-
12520
1       2  CA-2016-152156 2016-11-08 2016-11-11    Second Class    CG-
12520
2       3  CA-2016-138688 2016-06-12 2016-06-16    Second Class    DV-
13045
3       4  US-2015-108966 2015-10-11 2015-10-18  Standard Class    SO-
20335
4       5  US-2015-108966 2015-10-11 2015-10-18  Standard Class    SO-
20335

     Customer Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
2  Darrin Van Huff  Corporate  United States      Los Angeles  ...
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...

  Postal Code  Region       Product ID        Category Sub-
Category  \
```

```
0       42420   South  FUR-BO-10001798        Furniture    Bookcases

1       42420   South  FUR-CH-10000454        Furniture       Chairs

2       90036    West  OFF-LA-10000240  Office Supplies       Labels

3       33311   South  FUR-TA-10000577        Furniture       Tables

4       33311   South  OFF-ST-10000760  Office Supplies      Storage


                                        Product Name     Sales
Quantity  \
0                  Bush Somerset Collection Bookcase  261.9600
2
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
3
2  Self-Adhesive Address Labels for Typewriters b...   14.6200
2
3      Bretford CR4500 Series Slim Rectangular Table  957.5775
5
4                     Eldon Fold 'N Roll Cart System   22.3680
2

   Discount    Profit
0      0.00   41.9136
1      0.00  219.5820
2      0.00    6.8714
3      0.45 -383.0310
4      0.20    2.5164

[5 rows x 21 columns]
```

```
data.describe()
```

```
            Row ID   Postal Code         Sales     Quantity
Discount  \
count  9994.000000   9994.000000   9994.000000  9994.000000
9994.000000
mean   4997.500000  55190.379428    229.858001     3.789574
0.156203
std    2885.163629  32063.693350    623.245101     2.225110
0.206452
min       1.000000   1040.000000      0.444000     1.000000
0.000000
25%    2499.250000  23223.000000     17.280000     2.000000
0.000000
50%    4997.500000  56430.500000     54.490000     3.000000
0.200000
75%    7495.750000  90008.000000    209.940000     5.000000
```

```
0.200000
max    9994.000000  99301.000000  22638.480000    14.000000
0.800000

            Profit
count  9994.000000
mean     28.656896
std     234.260108
min   -6599.978000
25%       1.728750
50%       8.666500
75%      29.364000
max    8399.976000
```

Converting Date Column

```
data['Order Date']= pd.to_datetime(data['Order Date'])
data['Ship Date']= pd.to_datetime(data['Ship Date'])

data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   datetime64[ns]
 3   Ship Date      9994 non-null   datetime64[ns]
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: datetime64[ns](2), float64(3), int64(3), object(13)
memory usage: 1.6+ MB
```

```
data['Order Month']= data['Order Date'].dt.month
data['Order Year']= data['Order Date'].dt.year
data['Order Day of Week']= data['Order Date'].dt.dayofweek

data.head()
```

```
   Row ID        Order ID Order Date  Ship Date       Ship Mode
Customer ID  \
0       1  CA-2016-152156 2016-11-08 2016-11-11    Second Class    CG-
12520
1       2  CA-2016-152156 2016-11-08 2016-11-11    Second Class    CG-
12520
2       3  CA-2016-138688 2016-06-12 2016-06-16    Second Class    DV-
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3       4  US-2015-108966 2015-10-11 2015-10-18  Standard Class    SO-
20335
4       5  US-2015-108966 2015-10-11 2015-10-18  Standard Class    SO-
20335

     Customer Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
2  Darrin Van Huff  Corporate  United States      Los Angeles  ...
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...

          Category  Sub-Category  \
0        Furniture     Bookcases
1        Furniture        Chairs
2  Office Supplies        Labels
3        Furniture        Tables
4  Office Supplies       Storage

                                        Product Name     Sales
Quantity  \
0                  Bush Somerset Collection Bookcase  261.9600
2
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
3
2  Self-Adhesive Address Labels for Typewriters b...   14.6200
2
3      Bretford CR4500 Series Slim Rectangular Table  957.5775
5
4                     Eldon Fold 'N Roll Cart System   22.3680
2

  Discount    Profit  Order Month  Order Year  Order Day of Week
0     0.00   41.9136           11        2016                  1
1     0.00  219.5820           11        2016                  1
2     0.00    6.8714            6        2016                  6
```

```
3     0.45 -383.0310           10         2015                     6
4     0.20    2.5164           10         2015                     6

[5 rows x 24 columns]
```

Monthly Sales Analysis

```
sales_by_month= data.groupby('Order Month')
['Sales'].sum().reset_index()

sales_by_month

    Order Month        Sales
0             1   94924.8356
1             2   59751.2514
2             3  205005.4888
3             4  137762.1286
4             5  155028.8117
5             6  152718.6793
6             7  147238.0970
7             8  159044.0630
8             9  307649.9457
9            10  200322.9847
10           11  352461.0710
11           12  325293.5035

fig = px.line(sales_by_month,
             x='Order Month',
             y= 'Sales',
             title='Monthly Sales Analysis')
fig.show()
```

Sales by Category

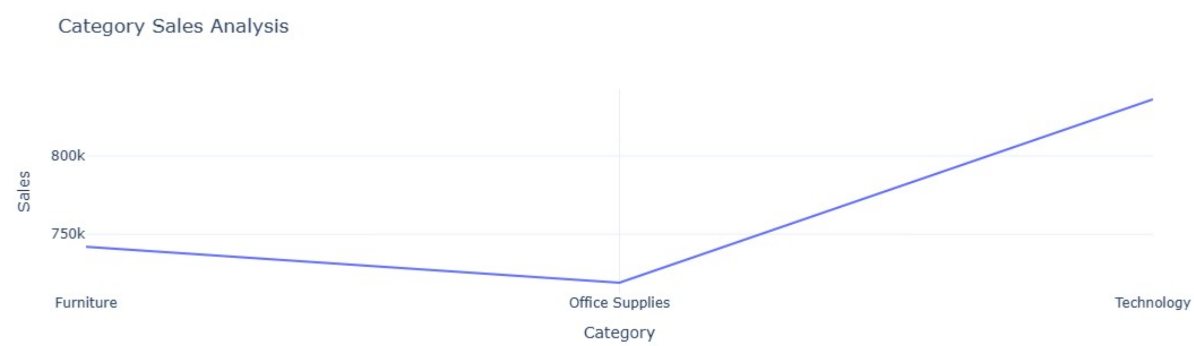
```
sales_by_category= data.groupby('Category')
['Sales'].sum().reset_index()
```

```
sales_by_category
```

	Category	Sales
0	Furniture	741999.7953
1	Office Supplies	719047.0320
2	Technology	836154.0330

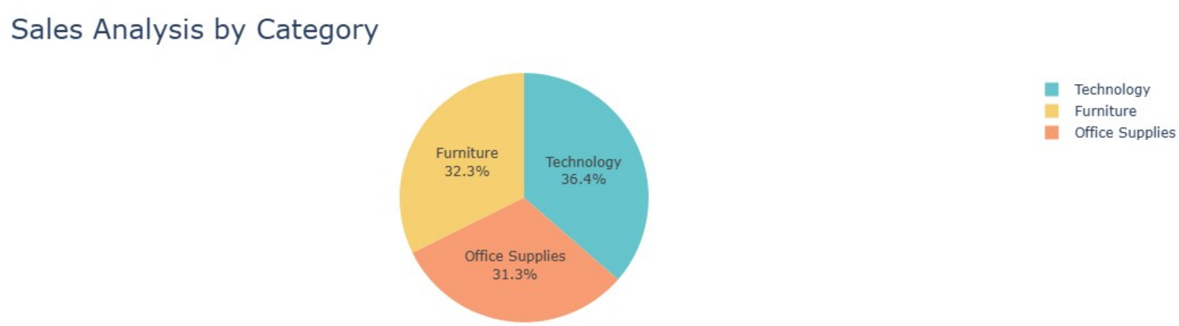
Sales by Category through graph

```
fig = px.line(sales_by_category,
             x='Category',
             y='Sales',
             title='Category Sales Analysis')
fig.show()
```



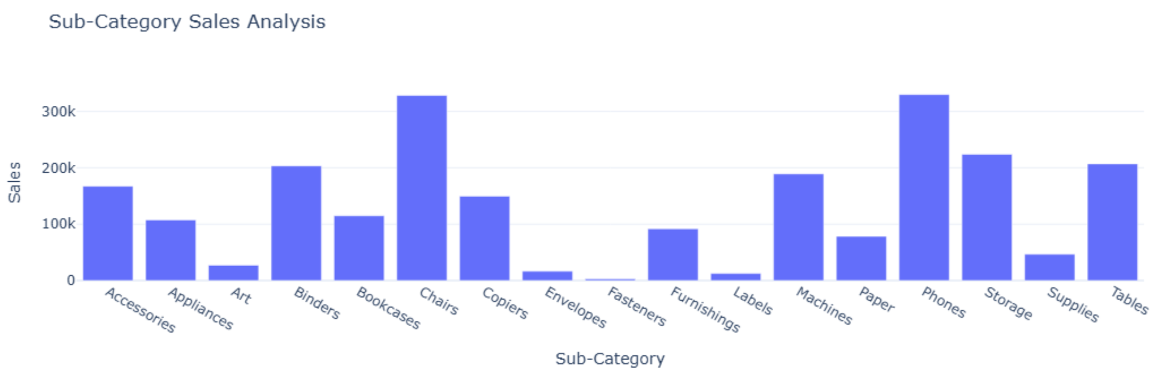
Sales by Category through Pie Chart

```
fig = px.pie(sales_by_category,
            values='Sales',
            names='Category',
            hole=0,
            color_discrete_sequence=px.colors.qualitative.Pastel)
fig.update_traces(textposition = 'inside', textinfo = 'percent+label')
fig.update_layout(title_text = 'Sales Analysis by Category',
title_font = dict(size=24))
fig.show()
```

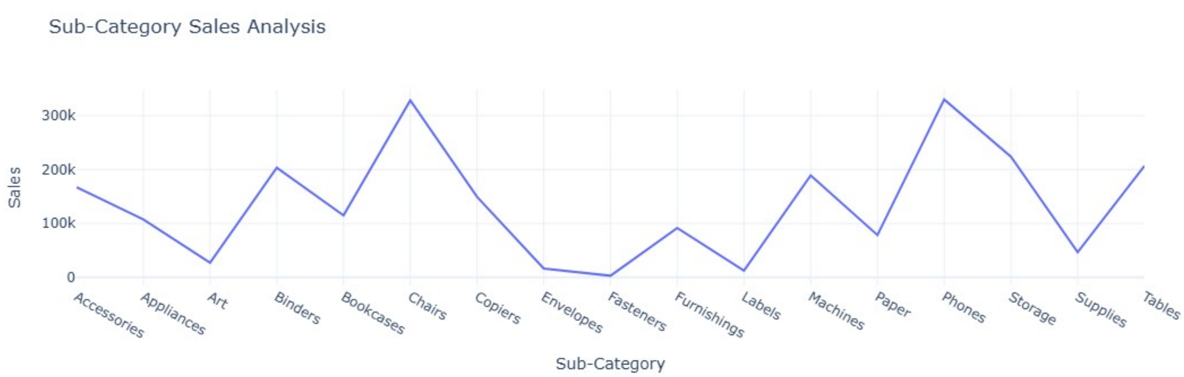


Sales Analysis by Sub-Category

```
sales_by_subcategory= data.groupby('Sub-Category')
['Sales'].sum().reset_index()

sales_by_subcategory

   Sub-Category        Sales
0   Accessories  167380.3180
1    Appliances  107532.1610
2           Art   27118.7920
3       Binders  203412.7330
4     Bookcases  114879.9963
5        Chairs  328449.1030
6       Copiers  149528.0300
7     Envelopes   16476.4020
8     Fasteners    3024.2800
9   Furnishings   91705.1640
10       Labels   12486.3120
11     Machines  189238.6310
12        Paper   78479.2060
13       Phones  330007.0540
14      Storage  223843.6080
15     Supplies   46673.5380
16       Tables  206965.5320

fig=px.bar(sales_by_subcategory,
          x='Sub-Category',
          y='Sales',
          title='Sub-Category Sales Analysis')
fig.show()
```




```
fig=px.line(sales_by_subcategory,
          x='Sub-Category',
          y='Sales',
          title='Sub-Category Sales Analysis')
fig.show()
```



profit by month

```
profit_by_month= data.groupby('Order Month')
['Profit'].sum().reset_index()

profit_by_month
```

	Order Month	Profit
0	1	9134.4461
1	2	10294.6107
2	3	28594.6872
3	4	11587.4363
4	5	22411.3078
5	6	21285.7954
6	7	13832.6648
7	8	21776.9384
8	9	36857.4753
9	10	31784.0413

```
10            11  35468.4265
11            12  43369.1919
```

```
fig= px.line(profit_by_month,
            x='Order Month',
            y='Profit',
            title= 'Monthly Profit Analysis')
fig.show()
```

Profit by Category

```
profit_by_category= data.groupby('Category')
['Profit'].sum().reset_index()

profit_by_category
```

	Category	Profit
0	Furniture	18451.2728
1	Office Supplies	122490.8008
2	Technology	145454.9481

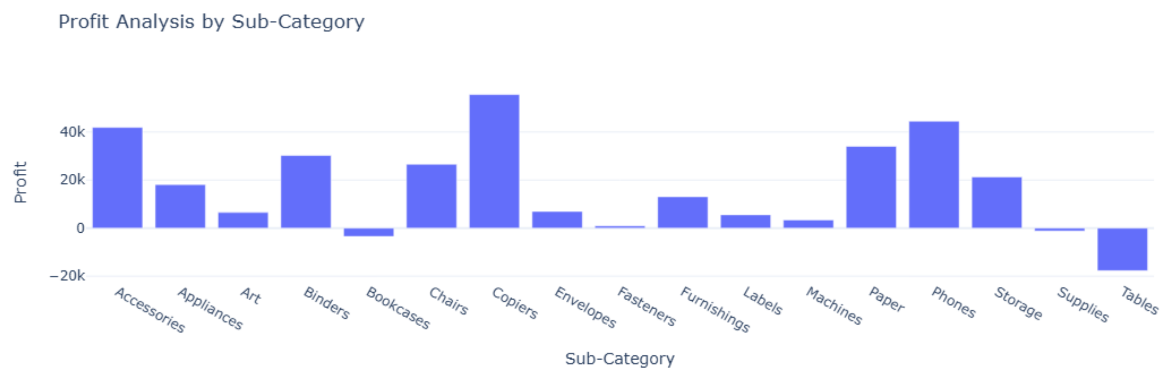
```
fig= px.pie(profit_by_category,
           values='Profit',
           names='Category',
           hole=0.5,
            color_discrete_sequence=px.colors.qualitative.Pastel)
fig.update_traces(textposition='inside', textinfo='percent+label')
fig.update_layout(title_text='Profit Analysis by Category',
title_font=dict(size=25))
fig.show()
```

Profit by Sub Category

```
profit_by_subcategory= data.groupby('Sub-Category')
['Profit'].sum().reset_index()

profit_by_subcategory
```

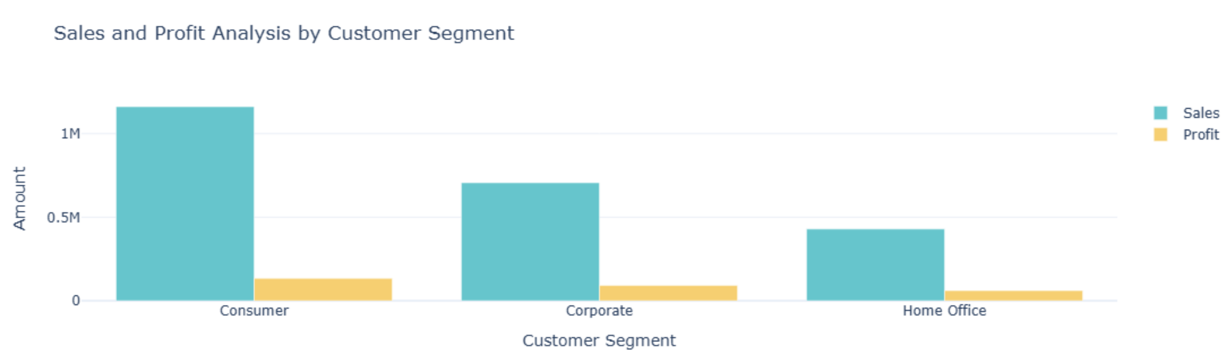
	Sub-Category	Profit
0	Accessories	41936.6357
1	Appliances	18138.0054
2	Art	6527.7870
3	Binders	30221.7633
4	Bookcases	-3472.5560
5	Chairs	26590.1663
6	Copiers	55617.8249
7	Envelopes	6964.1767
8	Fasteners	949.5182
9	Furnishings	13059.1436
10	Labels	5546.2540
11	Machines	3384.7569
12	Paper	34053.5693
13	Phones	44515.7306
14	Storage	21278.8264
15	Supplies	-1189.0995
16	Tables	-17725.4811

```
fig= px.bar(profit_by_subcategory,
            x='Sub-Category',
            y='Profit',
            title='Profit Analysis by Sub-Category')
fig.show()
```



Sales and Profit- Customer Segment

```
sales_profit_by_segment = data.groupby('Segment').agg({'Sales': 'sum',
'Profit': 'sum'}).reset_index()

color_palette = colors.qualitative.Pastel

fig = go.Figure()
fig.add_trace(go.Bar(x=sales_profit_by_segment['Segment'],
                     y=sales_profit_by_segment['Sales'],
                     name='Sales',
                     marker_color=color_palette[0]))

fig.add_trace(go.Bar(x=sales_profit_by_segment['Segment'],
                     y=sales_profit_by_segment['Profit'],
                     name='Profit',
                     marker_color=color_palette[1]))

fig.update_layout(title='Sales and Profit Analysis by Customer
Segment',
                  xaxis_title='Customer Segment',
yaxis_title='Amount')

fig.show()
```



Sales to Ratio

```
sales_profit_by_segment=data.groupby('Segment').agg({'Sales':'sum',
'Profit':'sum'}).reset_index()
sales_profit_by_segment['Sales_to_Profit_Ratio']=
sales_profit_by_segment['Sales']/sales_profit_by_segment['Profit']
print(sales_profit_by_segment[['Segment','Sales_to_Profit_Ratio']])
```

```
       Segment  Sales_to_Profit_Ratio
0     Consumer               8.659471
1    Corporate               7.677245
2  Home Office               7.125416
```